

Mohamed Abuella

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Scholar: Mohamed Abuella

Summary

Electrical engineer by training, traditionally interested in mathematical and computational analysis, modeling, and optimization, and data analytics for integrated energy systems planning and operations.

As a researcher, I focus on modernizing the electric grid and optimizing the integration of distributed energy resources by applying descriptive, predictive, and prescriptive analytics to improve energy efficiency and support the transition to net-zero infrastructures. My work includes recent applications in hydrogen and maritime shipping, with a broader interest in utilizing Artificial Intelligence to foster sustainability. Seeking to transfer and expand my expertise to bridge the gap between advanced research and real-world application.

Keywords: *Energy systems modeling and analytics, planning and operation, renewable energy and hydrogen integration, optimization and game theory, energy transition policy, net-zero pathways, AI/ML for energy forecasting and efficiency.*

🔗 For more details, see my online CV and professional profile in mohamedabuella.github.io/cv.

Education

University of North Carolina at Charlotte (UNCC)

USA

Ph.D in Electrical Engineering, GPA 4.0

2014–2018

Southern Illinois University at Carbondale (SIUC)

USA

M.Sc in Electrical and Computer Engineering, GPA 4.0

2010–2012

Higher Polytechnic Institute & College of Industrial Technology at Misurata

Libya

DipHE in Instrumentation 82%, and B.Tech Electromechanical Engineering, 86%

2002–2008

Experience

Research Fellow

UK

Northumbria University, Department of Mathematics, Physics and Engineering

May 2024–May 2026

Supporting the delivery of the “Hydrogen Integration for Accelerated Energy Transitions (HI-ACT)” research project.

See more details in [HI-ACT Project](#).

○ Acquired Expertise: *Integrated Energy Systems, Game Theory, Strategic & Policy Modeling, Hydrogen Integration, Energy Transition*

Researcher

Sweden

Halmstad University

April 2022–April 2024

Postdoctoral Researcher at the Center for Applied Intelligent Systems Research (CAISR). Research on AI for Sustainability applying Machine Learning techniques.

For more details in [iHelm Project](#).

○ Acquired Expertise: *Integrated Industry-Academia Collaboration, Big-Data & Spatiotemporal Analysis, Explainable AI (XAI), Research Methodologies & Project Supervision.*

Lecturer

Libya

College of Industrial Technology at Misurata

February 2020–March 2022

Taught Electrical Circuits, Electrical Measurements, Math 101.

○ Acquired Expertise: *Curriculum Revision & Preparing, Dedication, Listening, "Try to Modeling the Student's Way of Thinking".*

Research Assistant

USA

Energy Production and Infrastructure Center (EPIC) at UNC Charlotte

2014–2019

Statistical and Predictive Analytics to Modernize the Grid and Optimize its Integration of Renewables, focusing on Solar Energy Resources.

For more details in [Graduate Research Portfolio](#).

○ Acquired Expertise: *Energy Analytics, Energy Markets, Renewable Energy & Supply Chain, Machine Learning.*

M.Sc Research

USA

Department of Electrical and Computer Engineering at SIUC

2010–2012

Optimization for Electric Power Systems Including Wind Power.

For more details in [Graduate Research Portfolio](#).

○ Acquired Expertise: *Power Systems Analysis, Operation and Planning, Systems Optimization, Smart Grid.*

Teaching Assistant and Lab Instructor

College of Industrial Technology at Misurata

Taught Mathematics, Power Systems Analysis, and PLC.

○ Acquired Expertise: *Teaching, Tutorials, Lab Modeling & Simulations.*

Libya

2008–2009

Electrical Technician

Residential Electrical Wiring and Water & Wastewater Company

Wiring and maintain electrical control equipment.

○ Acquired Expertise: *Electrical Wiring & Installations, Maintenance & Operation.*

Libya

2000–2008

Recognitions

Best Paper Award (Third Prize): RPG 2025 – 14th Int. Conf. on Renewable Power Generation, Shanghai, China 2025

Outstanding Reviewer: IEEE Transactions on Sustainable Energy 2017

Third Prize for Student Papers: The 47th North American Power Symposium 2015

The Institute Prize for team of Energy Analytics Lab at UNCC: Global Energy Forecasting Competition 2014

The 1st Place: Department of Electromechanical Engineering at College of Industrial Technology 2008

Publications

Authored numerous peer-reviewed publications (all as first author). For the complete list, see my Google Scholar profile: [Mohamed Abuella](#). Accumulating over 750 citations and current H-index is **9**. my [ORCID](#).

Journal Articles

1. M. Abuella, A. Allahham, S. L. Walker, "A Policy-Navigation Framework for Exploring Hydrogen Integration Pathways in Great Britain Towards Net Zero," *Energy Policy*, vol. 218, Article 115535, Nov. 2026. DOI: [10.1016/j.enpol.2026.115535](#).
2. M. Abuella, A. Allahham, S. L. Walker, "Game Theory Approaches to Hydrogen Infrastructure Investment Planning in Great Britain: A Comparative Analysis of Competitive and Cooperative Frameworks," *International Journal of Hydrogen Energy*, vol. 220, p. 154064, 2026. DOI: [10.1016/j.ijhydene.2026.154064](#).
3. M. Abuella, A. Allahham, S. L. Walker, "A Cooperative Planning Framework for Hydrogen Blending in Great Britain's Integrated Energy System," *Energies*, vol. 19, no. 9, p. 2018, 2026. DOI: [10.3390/en19092018](#).
4. M. Abuella, A. Allahham, N. A. Rufa'I, S. L. Walker, "Hydrogen-Based Decarbonisation Strategies for Residential Heating: An Energy Efficiency and Conservation Analysis in the North of Tyne Region," *Energies*, vol. 18, no. 23, p. 6237, 2025. DOI: [10.3390/en18236237](#).
5. M. Abuella, H. Fanaee, S. Nowaczyk, S. Johansson, E. Faghani, "Time-Series Analysis Approach for Improving Energy Efficiency of Fixed-Route Passenger Vessel in Short-Sea Shipping," *Ocean Engineering*, vol. 334, p. 121555, 2025. DOI: [10.1016/j.oceaneng.2025.121555](#).
6. M. Abuella, M. Atoui, S. Nowaczyk, S. Johansson, E. Faghani, "Spatial Clustering Approach for Vessel Path Identification," *IEEE Access*, vol. 12, pp. 66248–66258, 2024. DOI: [10.1109/ACCESS.2024.3399116](#).
7. M. Abuella and B. Chowdhury, "Forecasting of Solar Power Ramp Events: A Post-Processing Approach," *Renewable Energy*, vol. 133, pp. 1380–1392, 2019. DOI: [10.1016/j.renene.2018.09.005](#).
8. M. Abuella and B. Chowdhury, "Improving Combined Solar Power Forecasts Using Estimated Ramp Rates: Data-Driven Post-Processing Approach," *IET Renewable Power Generation*, vol. 12, no. 10, pp. 1127–1135, 2018. DOI: [10.1049/iet-rpg.2017.0447](#).

Conference Papers.....

1. M. Abuella, A. Allahham, S. L. Walker, "Socio-Technical, Electrification, and Hydrogen-Driven Pathways for Residential Heating Decarbonisation in the North of Tyne," in *9th International Conference on Environment Friendly Energies and Applications (EFEA 2025)*, 4–5 December 2025, Newcastle upon Tyne, UK.
2. M. Abuella, A. Allahham, S. L. Walker, "A Game-Theoretic Framework for Hydrogen Integration to Accelerate Energy Transitions in Great Britain," in *14th International Conference on Renewable Power Generation (RPG 2025)*, 24–26 October 2025, Shanghai, China.
3. M. Abuella, M. Atoui, S. Nowaczyk, S. Johansson, E. Faghani, "Data-Driven Explainable Artificial Intelligence for Energy Efficiency in Short-Sea Shipping," in *ECML PKDD*, 2023.
4. M. Abuella and B. Chowdhury, "Adjusting Post-Processing Approach for Very Short-Term Solar PV Power Forecasts," in *2021 IEEE MI-STA*, 2021.
5. M. Abuella and B. Chowdhury, "Qualifying Combined Solar Power Forecasts in Ramp Events' Perspective," in *IEEE Power and Energy Society General Meeting*, 2018.
6. M. Abuella and B. Chowdhury, "Forecasting Solar Power Ramp Events Using Machine Learning Classification Techniques," in *IEEE PEDG*, 2018.
7. M. Abuella and B. Chowdhury, "Hourly Probabilistic Forecasting of Solar Power," in *North American Power Symposium (NAPS)*, 2017.
8. M. Abuella and B. Chowdhury, "Random Forest Ensemble of Support Vector Regression Models for Solar Power Forecasting," in *IEEE ISGT*, 2017.
9. M. Abuella and B. Chowdhury, "Solar Power Forecasting Using Support Vector Regression," in *ASEM International Annual Conference*, 2016.
10. M. Abuella and B. Chowdhury, "Solar Power Forecasting Using Artificial Neural Networks," in *North American Power Symposium (NAPS)*, 2015.
11. M. Abuella and B. Chowdhury, "Solar Power Probabilistic Forecasting by Using Multiple Linear Regression Analysis," in *IEEE SoutheastCon*, 2015.